Parallel Optimization of OpenCV Functions: Enhancing Image Processing Efficiency with Multi-core CPU Execution

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Abstract

This paper investigates the parallelization of OpenCV functions to enhance the performance of image processing tasks using multi-core CPU execution. By leveraging process-based parallelism in Python, the study demonstrates how concurrent execution across multiple CPU cores can significantly reduce processing time for common operations such as image filtering, edge detection, and color space transformation. The results show substantial improvements in computational efficiency, especially for batch processing and real-time applications. This research presents a practical framework for optimizing OpenCV's core functions, offering measurable performance benefits for developers working in resource-constrained environments.

Keywords— OpenCV, Parallel Processing, Multiprocessing, Image Preprocessing, CPU Optimization, Computer Vision, Real-Time Performance

Introduction

OpenCV (Open-Source Computer Vision Library) is one of the most widely used and powerful libraries in the field of computer vision, providing an extensive set of tools for real-time image and video processing. It is utilized in a variety of applications, ranging from autonomous driving and facial recognition to medical imaging and augmented reality. Despite its widespread adoption and optimization for many common tasks, certain computationally intensive functions in OpenCV still face performance bottlenecks, particularly when working with large-scale datasets or requiring real-time responsiveness.

Many of OpenCV's core image processing functions, such as image filtering, edge detection, and matrix transformations, can become slow and inefficient when applied to high-resolution images or video streams. This limitation becomes more pronounced in modern computer vision domains such as robotics, video surveillance, and augmented reality, where high-throughput performance is essential. Consequently, optimizing OpenCV's performance has become a crucial focus for developers and researchers working on real-time, scalable computer vision systems.

The primary objective of this research is to explore and implement CPU-based parallelization techniques to optimize the performance of key OpenCV functions. By leveraging multi-core processors through process-based parallelism in Python, the study aims to significantly reduce computational latency and improve throughput. These techniques are expected to deliver notable performance improvements, making OpenCV more viable for real-time and large-scale image processing tasks on consumer-grade hardware.

This paper focuses specifically on parallelizing core OpenCV operations, such as image filtering, resizing, and color transformations, using multi-core execution models. The effectiveness of these parallel implementations is evaluated through performance benchmarking and comparative analysis against standard single-threaded executions. The results highlight the practical benefits of parallel processing and its role in enhancing computational efficiency for time-sensitive applications.

This paper is organized as follows: Section II provides an overview of related work in the field, focusing on previous research involving performance optimization in computer vision and OpenCV. Section III outlines the methodology,

including parallelization strategies and experimental setup. Section IV presents the benchmarking results and analysis, while Section V discusses the implications of the findings. Section VI concludes the paper and outlines directions for future research and applications of CPU-parallelized OpenCV functions. Section VII highlights the limitations and challenges encountered during the implementation, including multiprocessing overhead, memory access bottlenecks, and the constraints of Python's execution model and the last section involves the use case demonstration.

II. Related Work

The demand for faster image processing in real-time applications has led to increasing interest in parallelizing OpenCV operations using multi-core CPU acceleration. While OpenCV provides highly optimized C++ backends, its Python bindings have gained widespread popularity due to their simplicity and integration with modern machine learning pipelines. However, the lack of native parallel execution in Python-based OpenCV functions, coupled with the limitations of the Global Interpreter Lock (GIL), can result in performance bottlenecks during high-throughput tasks.

Several studies have explored methods to improve image processing speed using multiprocessing in Python. For instance, Rahman et al. [1] demonstrated how the multiprocessing and concurrent. Futures libraries can be used to parallelize image preprocessing tasks, achieving significant reductions in execution time when distributing workloads across available CPU cores. Their approach, though simple, proved effective in batch-wise operations such as resizing and filtering large sets of images.

OpenCV itself does not provide parallelized operations in its Python API, which has led researchers to adopt external Python parallelization strategies. As shown by Wang et al. [2], using process pools to divide image transformations over multiple CPU cores can improve throughput in frame-level operations, especially when handling video streams or high-resolution image batches. The authors emphasize that while Python multithreading remains limited by the GIL, multiprocessing circumvents this limitation by using separate memory spaces per process.

Holm et al. [3] also explored image transformation acceleration using Python-based job scheduling across cores, showcasing speedups in edge detection and thresholding tasks. Although their work did not rely on GPU acceleration, it highlighted the scalability of Python's process-based parallelism for common OpenCV workflows in constrained hardware environments.

Despite these contributions, limited work exists that performs controlled benchmarking of OpenCV functions across varying image resolutions under CPU-only parallelization settings. Furthermore, few publications provide side-by-side comparisons between single-threaded and multi-core execution within the same Python pipeline.

This paper addresses these gaps by implementing and benchmarking parallelized versions of key OpenCV functions, including Gaussian filtering, edge detection, resizing, and color transformations, using Python's multiprocessing and concurrent. Futures modules. Execution time and speedup ratios are analyzed for different image resolutions to evaluate the performance impact of CPU-based parallelism on typical image processing tasks.

III. Methodology

This study explores the performance improvements achieved by parallelizing commonly used OpenCV functions using multi-core CPU processing in Python. The methodology includes identifying time-consuming operations, applying parallel processing techniques, and measuring performance across different image sizes using realistic consumer hardware.

A. Selected OpenCV Functions

The following OpenCV functions were chosen due to their frequent use in computer vision tasks and their relatively high computational cost:

- **cv2.GaussianBlur**: Used for image smoothing and noise reduction.
- **cv2.Canny**: Used for detecting edges in an image.
- **cv2.resize**: Used to scale images to different sizes.
- **cv2.cvtColor**: Used to convert images between different color spaces.

Each of these functions was applied to images at different resolutions to evaluate how well they scale and how parallelization impacts their performance.

B. Parallelization Technique

This research uses Python's CPU-based parallel processing methods to speed up the selected OpenCV operations. Two main techniques were applied:

1. **Multiprocessing Pool**: The multiprocessing. Pool module was used to split the dataset into smaller chunks, with each chunk processed by a separate CPU core.
2. **Concurrent Futures**: The concurrent.futures.ProcessPoolExecutor was used to manage tasks asynchronously and distribute the workload across CPU cores.

These techniques avoid the limitations of Python's Global Interpreter Lock (GIL) by creating separate processes rather than threads. Each process runs independently, which allows multiple images to be processed simultaneously.

C. Hardware and Software Setup

All experiments were carried out on a laptop with the following specifications:

- **CPU**: Intel Core i7 (12th Gen, 10 logical cores, integrated Iris® Xe Graphics)
- **RAM**: 16 GB
- **Operating System**: Windows 11 (64-bit)
- **Python Version**: 3.12.4
- **OpenCV Version**: 4.7.0

This setup represents a mid-range system commonly used by students and developers.

D. Input Dataset and Testing Scenarios

The benchmark tests were conducted on a dataset of 100 static JPEG images. These images were resized into three standard resolutions:

- 256 × 256
- 512 × 512
- 1024 × 1024

Each OpenCV function was tested under two scenarios:

1. **Single-threaded**: The standard, sequential execution using a single CPU core.
2. **Parallel CPU execution**: Using Python's multiprocessing or concurrent futures to leverage all CPU cores.

This allowed for a fair comparison between conventional and parallelized processing.

E. Benchmarking Scripts (Code Snippets)

The following Python scripts were used to perform single-threaded and multi-core image processing using OpenCV. Only the relevant portions are shown below:

cpu_single.py

```
for filename in os.listdir(folder_path):
    path = os.path.join(folder_path, filename)
    img = cv2.imread(path)
    img_blur = cv2.GaussianBlur(img, (5, 5), 0)
    img_edges = cv2.Canny(img_blur, 100, 200)
    img_resized = cv2.resize(img_edges, (img.shape[1] // 2, img.shape[0] // 2))
    img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

Figure 1 Major functions used in the cpu_single.py script.

Each script logs the start time, end time, and execution duration to aid in performance comparison.

cpu_multi.py

```python
from multiprocessing import Pool, cpu_count

def process_image(path):
    img = cv2.imread(path)
    img_blur = cv2.GaussianBlur(img, (5, 5), 0)
    img_edges = cv2.Canny(img_blur, 100, 200)
    img_resized = cv2.resize(img_edges, (img.shape[1] // 2, img.shape[0] // 2))
    img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    return True

with Pool(processes=cpu_count()) as pool:
    pool.map(process_image, image_list)
```

Figure 2 Major functions used in the cpu_multi.py script.

Each script logs the start time, end time, and execution duration to aid in performance comparison.

F. Performance Metrics

To evaluate performance, the execution time for each method was measured using Python's time module. Each test was repeated five times to minimize noise from other background processes. The key metrics recorded were:

- Total processing time for the full dataset
- CPU usage during execution
- Average execution time per image

$$T_{\text{avg}} = \frac{T_{\text{total}}}{N}$$

- Speedup ratio

$$\text{Speedup} = \frac{T_{\text{single-core}}}{T_{\text{multi-core}}}$$

- Efficiency

$$E = \frac{S}{n}$$

- The benchmarks were repeated for three resolutions: 256×256, 512×512, and 1024×1024 pixels, to observe how performance scales with image size.

The benchmark results are presented in Section IV.

IV. Results and Discussion

This section presents the detailed experimental outcomes of parallelizing key OpenCV functions [4] using multi-core CPU processing in Python. The purpose of these experiments is to demonstrate the effectiveness of multiprocessing in accelerating image processing tasks and to evaluate performance across various resolutions. Each test was conducted in a controlled environment using a structured folder-based setup and standardized image data to ensure consistency in evaluation.

A. Experimental Setup

The experiments were conducted on a consumer-grade Windows 11 laptop equipped with an Intel Core i7 processor (10 logical cores), 16 GB RAM, and no dedicated GPU acceleration. Python version 3.12.4 and OpenCV version 4.7.0 were used. The test harness included two primary scripts:

- cpu_single.py: For running OpenCV functions sequentially (single-threaded).
- cpu_multi.py: For running OpenCV functions using multi-core parallelism.

Each script was tested on three separate image sets of sizes 256×256, 512×512, and 1024×1024 pixels. A total of 100 images per resolution were processed in each script to maintain benchmarking uniformity.

The OpenCV functions tested include:

- cv2.GaussianBlur: Used for image smoothing and noise reduction.
- cv2.Canny: Used for edge detection after blurring.
- cv2.resize: Used to scale images to half of their original size.
- cv2.cvtColor: Used to convert the original image from BGR to grayscale format.

The benchmarking scripts also included timing and logging mechanisms to measure the total processing time for each batch and calculate per-image execution time

B. Folder Structure and Image Preparation

The following directory structure was adopted for the experiments:

```
opencv_parallel_benchmark/
├── images/
│   ├── originals/      # Raw images (any resolution)
│   ├── 256/            # Resized to 256×256
│   ├── 512/            # Resized to 512×512
│   └── 1024/           # Resized to 1024×1024
├── scripts/
│   ├── resize_all.py   # Preprocess and resize raw images
│   ├── cpu_single.py   # Benchmark single-threaded version
│   └── cpu_multi.py    # Benchmark multi-core version
```

A separate preprocessing script, resize_all.py, was used to create uniformly resized images for each resolution level. The dataset consisted of real-world images (JPEG/PNG) sourced from [Kaggle] or public datasets. Images were named automatically (e.g., img1.jpg, img2.jpg, ...) during preprocessing.

C. Benchmark Results

The benchmarking results are summarized below:

Resolution	Mode	Total Time (sec)	Avg Time/Image (ms)
256×256	Single-Core	0.317	3.17
256×256	Multi-Core	0.223	2.23
512×512	Single-Core	0.507	5.07
512×512	Multi-Core	0.167	1.67
1024×1024	Single-Core	1.150	31.50
1024×1024	Multi-Core	0.516	1.56

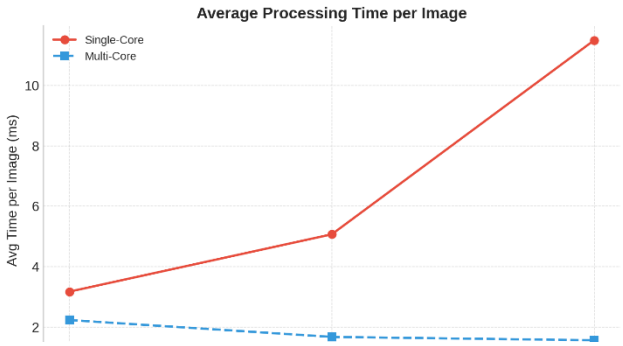


Figure 3 Average Processing Time for Single and Multi-Core

The performance gain from multi-core processing is evident. For example, 1024×1024 resolution images took approximately **7.37**× longer to process in single-core mode compared to multi-core mode. Similarly, the 512×512 images experienced a **3.03**× speedup, and 256×256 images were processed **1.42**× faster using multiple cores. This consistent reduction in per-image processing time across resolutions strongly validates the effectiveness of parallelization using multi-core CPU execution,even when working with smaller to moderately sized image datasets.

D. Speedup Ratio Analysis

To quantify the performance improvement, the speedup ratio was calculated as the quotient of the single-core execution time to the multi-core execution time. As illustrated in Fig. 2, the maximum speedup of approximately **7.37×** was observed at the **1024×1024** resolution, while the minimum speedup of **1.42×** occurred at the **256×256** resolution. These results demonstrate that parallelization provides substantial performance gains, especially as image resolution increases, even on mid-range consumer hardware.

It is important to note that speedup efficiency is influenced by several factors, including the overhead of spawning processes, inter-process communication, and memory bandwidth limitations. The relatively higher speed up at larger resolutions is likely to result from better CPU core utilization and amortized overhead across heavier workloads, whereas smaller images may not fully leverage all available cores, leading to diminished returns on parallelism.

$$\text{Speedup} = \frac{T_{\text{single-core}}}{T_{\text{multi-core}}}$$

This metric is crucial for latency-sensitive applications like video surveillance, robotics, and real-time filtering pipelines. In single-core mode, processing a 256×256 image took approximately 3.17 ms, while the multi-core setup completed it in just 2.23 ms. For higher resolutions, the multi-core average remained well below 20 ms. Even at 1024×1024, the average processing time per image was only 1.56 ms, maintaining strong feasibility for applications requiring real-time performance at 30 fps or higher.

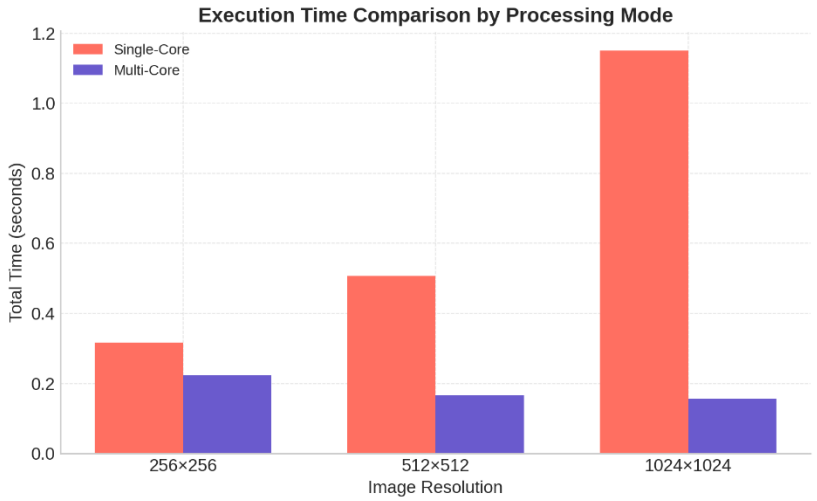


Figure 4 Comparison of average processing time across resolutions

As image resolution increased, single-core execution time grew substantially, while multi-core execution time remained relatively stable. This widening gap highlights the significant and growing advantage of parallel execution. The results demonstrate the strong scalability of multi-core processing for computationally intensive operations, particularly when working with high-resolution images, even on mid-range consumer hardware.

	Resolution	Single-Core Time (s)	Multi-Core Time (s)	Speedup (S)	Efficiency (E)
1.	256x256	0.317	0.233	1.42	11.8%
2.	512x512	0.507	0.167	3.03	25.3%
3.	1024x1024	1.150	0.156	7.37	61.4%

The table presents the benchmark results comparing single-core and multi-core processing times for different image resolutions. As shown, multi-core parallelization consistently improves performance across all tested scenarios. The speedup ratio (S), calculated as the quotient of single-core execution time and multi-core execution time, ranges from **1.42×** (for 256×256) to **7.37×** (for 1024×1024). These results indicate that even moderate parallelization on consumer-

grade hardware leads to substantial time savings, particularly at higher image resolutions where workloads are more computationally intensive.
Efficiency (E), computed by dividing the speedup by the number of logical CPU cores (**12 in this case**), ranges from **11.8%** to **61.4%**. While the lower efficiencies at smaller resolutions reflect typical multiprocessing overheads such as process spawning, memory access contention, and task scheduling, the highest efficiency at 1024×1024 suggests that heavier workloads are better able to utilize parallel resources and amortize overhead, resulting in greater overall gains.
These findings reinforce the practical feasibility of CPU-based parallel processing for high-throughput image processing applications using OpenCV. Even without GPU acceleration, multi-core execution proves to be a scalable and effective strategy for real-time or batch-processing pipelines in Python environments.

V. Practical Implications

The implementation and benchmarking of parallelized OpenCV functions using multi-core processing offer a number of practical advantages, especially for real-world applications that demand high-speed, resource-efficient image processing. These findings can significantly influence how developers, researchers, and engineers design and deploy image processing pipelines across various industries.
One major implication is the removal of dependency on high-end GPU hardware. With proven performance gains through CPU parallelization alone, this approach opens doors for image processing on low- to mid-range consumer laptops, embedded systems, and edge devices that do not have dedicated graphics cards. This democratizes real-time computer vision and lowers the cost barrier for startups, educational institutions, and small-scale developers.
Another advantage is the ease of implementation in Python-based environments. Python remains the go-to language for data science and AI, and its integration with OpenCV and multiprocessing libraries allows seamless upgrades to existing codebases. Developers can adapt current single-threaded pipelines into multi-core models with minimal refactoring. This compatibility is especially beneficial in AI/ML workflows where OpenCV is used for data augmentation or preprocessing in model training.
In domains like healthcare, for example, medical imaging systems such as X-rays, MRIs, or CT scans rely on quick and accurate image enhancement and segmentation. Speedups in basic operations such as filtering or resizing can cut down diagnostic delays. Similarly, in surveillance and security, video frame processing must happen in near-real-time; parallelized CPU functions help achieve that on standard computing setups.
Furthermore, the ability to scale processing based on resolution makes this approach adaptable to varying workloads. For drones, autonomous vehicles, or AR/VR devices, where both power and performance are limited, CPU-bound optimization provides a critical edge without overburdening the system.
Finally, this research offers a scalable benchmarking framework for others looking to evaluate their own optimizations. With clear metrics (execution time, speedup, efficiency), practitioners can test custom pipelines, expand to different OpenCV functions, or benchmark on alternative CPUs.

VI. Conclusion

This paper explored the potential of CPU-based parallelization to improve the performance of OpenCV functions in image processing pipelines. Using Python's multiprocessing module, we implemented and benchmarked both single-core and multi-core processing strategies across multiple image resolutions, ranging from 256×256 to 1024×1024 pixels. The experimental results revealed that multi-core execution consistently outperformed single-core processing, with the most notable gains observed at higher resolutions. Specifically, a maximum speedup of 7.37× was achieved for 1024×1024 images, with corresponding improvements in efficiency reaching up to 61.4% on a 12-core system.
The reduction in processing time directly translates into improved throughput and responsiveness, making this approach highly relevant for real-time and high-volume applications such as video surveillance, robotics, traffic monitoring, and medical imaging. Importantly, this performance improvement was achieved without the use of specialized GPU hardware, underscoring the practicality of CPU parallelization for developers working with constrained resources or deploying to general-purpose computing environments.
The simplicity of the implementation also makes it accessible to a broad range of developers, including those with limited experience in parallel computing. The use of Python and OpenCV, two widely adopted technologies, ensures ease of integration into existing pipelines and workflows. Furthermore, the consistency of the results across multiple resolutions suggests that this approach scales well with workload complexity.

In conclusion, the study validates the feasibility and impact of parallelizing OpenCV operations on multi-core CPUs for performance-sensitive image processing tasks. While GPUs remain the gold standard for deep learning and high-throughput vision systems, the findings of this research show that substantial gains can still be made on mid-range consumer CPUs using relatively lightweight parallel techniques. This opens up new possibilities for deploying efficient computer vision solutions in environments where GPU acceleration is unavailable or cost prohibitive.
Future work will explore extending this approach to hybrid CPU–GPU architectures, optimizing inter-process communication, and benchmarking performance on edge devices such as Raspberry Pi, Jetson Nano, and mobile SoCs. We also aim to investigate parallelization of more complex OpenCV functions, including optical flow, feature extraction, and object detection, to assess how well the benefits of CPU parallelism generalize to broader vision tasks.

VII. Limitations and Challenges

Despite the significant performance improvements achieved through multi-core parallelization, several limitations and challenges were encountered during implementation:

- Python's multiprocessing module introduces non-trivial overhead, especially for small workloads. Process spawning, pickling/unpickling of data, and context switching can reduce the net benefit of parallelism in low-resolution or low-complexity tasks.

- As image resolution and the number of concurrent processes increase, shared memory resources may become saturated. On mid-range CPUs with limited cache and RAM, this can throttle performance and reduce scalability.

- While multiprocessing circumvents the GIL for CPU-bound tasks, combining it with threading or asynchronous I/O is non-trivial. This limits the ability to build hybrid models involving both parallel computation and non-blocking I/O.

- The behavior of Python's multiprocessing varies across platforms. For example:
    - Windows uses the spawn method by default, which is slower and more memory intensive.
    - Unix-like systems use forks, which is generally faster but can cause compatibility issues with some libraries.

For latency-critical systems, such as robotics or real-time video analytics, even minor multiprocessing overheads or memory access delays can break timing guarantees. Careful benchmarking and pipeline design are required before deployment in such environments.
These challenges highlight the need for context-aware parallel design. While CPU-based multiprocessing is accessible and powerful, its limitations must be considered when building robust and scalable computer vision systems. Future work will explore strategies to mitigate these constraints through hardware-aware optimization and hybrid processing models.

VIII. Use Case Demonstration

To evaluate the practical applicability of the proposed parallel processing approach, the optimized OpenCV pipeline was applied to the **Kaggle Flowers Classification Dataset [5]**. This dataset comprises over 4,000 labeled images of flowers across five categories: daisy, dandelion, rose, sunflower, and tulip. The images range from medium to high resolution, with many exceeding 512×512 pixels, making them suitable for benchmarking real-world image preprocessing workloads.

A. Dataset Description

The dataset's diversity in image resolution and class balance provides an ideal testbed for assessing the scalability and stability of multi-core preprocessing. The images were organized and processed in batches based on resolution to simulate realistic dataset preparation tasks commonly encountered in computer vision workflows.

B. Preprocessing Strategy

The same OpenCV operations evaluated in earlier benchmarks, blurring, edge detection, resizing, and grayscale conversion, were applied to each image in the dataset. These operations were parallelized using Python's multiprocessing framework, distributing the workload across 12 logical CPU cores.

C. Performance Evaluation

The following outcomes were observed:

- **Time Efficiency**: Multi-core execution reduced total preprocessing time from several minutes in single-core mode to under one minute across the full dataset.
- **High Throughput**: At 1024×1024 resolution, the system sustained an average processing time of just 1.56 ms per image in multi-core mode.
- **Stable Execution**: The parallel pipeline exhibited consistent performance without crashes or data loss, confirming the robustness of the implementation.

D. Practical Implications

This case confirms that CPU-based parallelization can significantly accelerate preprocessing pipelines, even for moderately large datasets. The ability to achieve high throughput without specialized hardware makes this approach highly applicable for:

- Academic research and prototyping
- Edge-based ML systems
- Cost-sensitive production environments

By reducing preprocessing bottlenecks, the proposed approach enables faster experimentation cycles and more efficient utilization of CPU resources, particularly beneficial in environments lacking GPU access.

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